

Probabilistic Machine Learning

Lecture 04: Independence, Naive Bayes and Graphical Models

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Module 1

Bayes as a Classifier

Guiding Question: *The data tells us how attributes behave given the class. We need the reverse. What does the conversion cost?*

Thinking With Uncertainty

Bayes' theorem lets us update a **hypothesis** H in the light of **data** D :

$$P(H \mid D) = \frac{P(H) \cdot P(D \mid H)}{P(D)}$$

This is the whole apparatus of evidence-based decision making in one line: a belief, a measurement of how well the belief explains what we saw, and a normalisation.

Example: Do I Have the Flu?

You have a headache and a cough. From the literature:

Likelihood

90% of people with the flu
show headache and coughing

$$P(D \mid H) = 0.9$$

Prior

5% of people currently have
the flu

$$P(H) = 0.05$$

Evidence

headache and coughing occur
in 20% of people

$$P(D) = 0.2$$

How likely is it that you have the flu?

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How likely is it that you have the flu?

$$P(H \mid D) = \frac{0.05 \cdot 0.9}{0.2} = 0.225$$

Why So Low?

Few people have the flu

The prior $P(H) = 0.05$ caps how far the evidence can carry us.

The symptoms are common

$P(D) = 0.2$ means many people have these symptoms *without* the flu.

A symptom is informative to the degree that it is **rare in general** but **common under the hypothesis**. A likelihood alone tells you nothing.

Takeaway

Bayesian decision making weighs **prior × likelihood** against how ordinary the evidence is. Deciding under uncertainty means keeping all three numbers in view, not just the one the study reports.

Two Directions of the Same Connection

Observed connection

$$A_1, \dots, A_p \mid B$$

Given the class, how do the attributes behave? — the standard situation, and the one data can answer directly.

Reversed connection

$$B \mid A_1, \dots, A_p$$

Given the attributes, which class? — the diagnosis situation, and the one we actually need.

$$P(B \mid A_1, \dots, A_p) = \frac{P(B) \cdot P(A_1, \dots, A_p \mid B)}{P(A_1, \dots, A_p)}$$

What the Classifier Needs

Class priors

$$P(\text{class} = y)$$

How common is each class, before seeing any attribute?

Attribute likelihoods

$$P(\text{attribute} = v \mid \text{class} = y)$$

How does each class distribute its attribute values?

Given both, the posterior $P(\text{class} \mid \text{attributes})$ follows from Bayes — and the prediction is the class with the highest posterior.

Estimating $P(A_1, \dots, A_p \mid B)$ requires the **joint** distribution of all attributes within each class: exponentially many parameters, and never enough data.

Module 2

Naive Bayes

Guiding Question: *Its central assumption is almost always false, and it classifies well anyway. How can both be true?*

Generative and Discriminative Classifiers

Generative

Model the **joint** distribution $P(A, B)$ — how the data was produced — then derive $P(B | A)$.

Naive Bayes.

Discriminative

Model $P(B | A)$ **directly** — only the boundary between classes.

Logistic regression.

The Naive Assumption

Assume the attributes are **conditionally independent given the class**:

$$P(A_1, \dots, A_p \mid B) = \prod_{j=1}^p P(A_j \mid B)$$

Exponentially many parameters collapse to p per class — one distribution per attribute.

The assumption is almost always false. Naive Bayes nonetheless classifies well, because ranking the classes correctly is easier than estimating their probabilities correctly.

The Classifier

$$P(B_i \mid A_1, \dots, A_p) = \frac{P(B_i) \cdot \prod_{j=1}^p P(A_j \mid B_i)}{\sum_{i=1}^k P(B_i) \cdot \prod_{j=1}^p P(A_j \mid B_i)}$$

Score each class

prior × product of attribute likelihoods

Normalise

divide by the total over all classes — the theorem of total probability

Predict

the class with the largest posterior

For prediction alone the denominator can be skipped — it is the same for every class. It is needed only when the probability itself is reported.

Example: Enjoy Sport?

Outlook	Temperature	Humidity	Wind	EnjoySport
sunny	hot	high	weak	no
sunny	hot	high	strong	no
overcast	hot	high	weak	yes
rain	mild	high	weak	yes
rain	cool	normal	weak	yes
rain	cool	normal	strong	no

Estimate the priors and the per-attribute likelihoods by **counting**, then score a new day.

The Zero-Frequency Problem

If an attribute value never occurs with a class in the training data, its estimated likelihood is **zero** — and one zero factor drives the whole product to zero, whatever the other attributes say.

Laplace smoothing adds a pseudo-count to every combination, so that unseen does not mean impossible.

One missing count

vetoed every other piece of evidence

Naive Bayes buys tractability with a **conditional independence assumption it knows to be false** — and works anyway, because classification needs the right argmax, not the right probability.

Module 3

Independence, Properly

***Guiding Question:** Naive Bayes assumed independence to survive. What does the assumption actually claim — and when does conditioning create dependence rather than remove it?*

Conditional Independence

Definition. Events A and B are **conditionally independent given C** (written $A \perp\!\!\!\perp B \mid C$) iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C)$$

equivalently

$$P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B carries no further information about A .**

Dependent, but Independent Given C

Let $P(A) = \frac{8}{16}$, $P(B) = \frac{3}{16}$, $P(C) = \frac{4}{16}$ on a 16-cell sample space.

Unconditionally dependent

$$P(A \cap B) = \frac{1}{16}$$

$$P(A) \cdot P(B) = \frac{3}{32}$$

$$\frac{1}{16} \neq \frac{3}{32}, \text{ so } A \not\perp B$$

Conditionally independent given C

$$P(A \cap B \mid C) = \frac{1}{4}$$

$$P(A \mid C) \cdot P(B \mid C) = \frac{2}{4} \cdot \frac{2}{4} = \frac{1}{4}$$

$$\text{so } A \perp\!\!\!\perp B \mid C$$

Also directly: $P(A \mid B \cap C) = \frac{1}{2} = P(A \mid C)$.

Why This Matters

Explaining away

A common cause C can make two effects look related.

Model size

Conditional independence lets a joint distribution factorise into small pieces.

Later in the course

Naive Bayes assumes it.
Graphical models are built from it.

Conditional independence neither implies nor is implied by ordinary independence. Both directions can fail.

Takeaway

Conditioning can **create** independence as easily as it destroys it. "Given what I already know, does this extra fact still tell me anything?" is the question that makes large probabilistic models tractable.

Module 4

From Assumption to Model

Guiding Question: *Naive Bayes assumes everything is independent given the class. What if we could state exactly which independences hold?*

The Problem: Joint Distributions Explode

Consider a joint distribution $p(x_1, \dots, x_n)$ over n **binary** random variables. How many configurations are there?

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2^n states, $2^n - 1$ free parameters

Specification

For large n , writing down all values is impossible.

Inference

A single marginal $p(x_1)$ requires summing over 2^{n-1} states.

Chain rule?

It rewrites the joint exactly — but does not reduce the number of values.

What a Graphical Model Is

Definition (Graphical model). A graphical model depicts the dependence and independence relations of a distribution using a graph G as its underlying structure.

Modelling

Describe the variables of a domain and their interactions — how $p(x_1, \dots, x_n)$ **factorises**.

Inference

Answer queries such as $p(x_1 \mid x_2 = a, \dots)$ efficiently over that factorisation.

Usually one class of model is used for modelling and another for inference the latter is often compiled from the former.

How a Graph Can Help

Probabilistic inference is summing and multiplying. Let every variable be a node, and read edges as interaction:

No edge

$$f(a, b) = g(a) \cdot h(b)$$

The variables are independent — the factor splits.

Edge

$$f(a, b) \neq g(a) \cdot h(b)$$

The variables are dependent — the factor does not split.

Edge **direction** can encode further structure. The principle extends to any number of variables.

Graphical models have limited expressive power: not every conditional independence statement can be drawn, and different graphs can represent the same distribution.

Example: Why Is the Grass Wet?

Tracey leaves her house and sees that her grass is wet. **Was it rain, or did she leave the sprinkler on?** She notices her neighbour Jack's grass is also wet.

R

raining (1) or not (0)

S

Tracey's sprinkler on
(1) or off (0)

J

Jack's grass wet (1)
or dry (0)

T

Tracey's grass wet
(1) or dry (0)

Building the Model

The full joint over four binary variables needs $2^4 - 1 = 15$ probabilities. The chain rule rewrites it but does not shrink it:

$$p(T, J, R, S) = p(T \mid J, R, S)p(J \mid R, S)p(R \mid S)p(S)$$

Now bring in **background knowledge** — assumptions of conditional independence:

Tracey's grass does not depend on Jack's

$$p(T \mid J, R, S) = p(T \mid R, S)$$

Jack's grass does not depend on Tracey's sprinkler

$$p(J \mid R, S) = p(J \mid R)$$

Rain is not influenced by the sprinkler

$$p(R \mid S) = p(R)$$

$$p(T, J, R, S) = p(T \mid R, S)p(J \mid R)p(R)p(S)$$

Only $4 + 2 + 1 + 1 = 8$ probabilities remain — and each one is a question a person can actually answer.

The Probability Tables

Factor	$p(\cdot)$	$1 - p(\cdot)$
$p(R = 1)$	0.2	0.8
$p(S = 1)$	0.1	0.9
$p(J = 1 \mid R = 0)$	0.2	0.8
$p(J = 1 \mid R = 1)$	1.0	0.0
$p(T = 1 \mid R = 1, S = 0)$	1.0	0.0
$p(T = 1 \mid R = 1, S = 1)$	1.0	0.0
$p(T = 1 \mid R = 0, S = 1)$	0.9	0.1
$p(T = 1 \mid R = 0, S = 0)$	0.0	1.0

Missing edges in the graph are exactly the assumptions above the tables supply the numbers.

Inference: What Does the Evidence Say?

Rain explains the wetness — the sprinkler falls back towards its prior



Explaining Away

$$p(S = 1 \mid T = 1) = \frac{\sum_{J,R} p(T=1 \mid R, S=1)p(J \mid R)p(R)p(S=1)}{\sum_{J,R,S} p(T=1 \mid R, S)p(J \mid R)p(R)p(S)}$$

Two independent causes (R and S) become **dependent once their common effect is observed**. Learning that rain occurred lowers the probability of the sprinkler, even though rain and sprinkler are unconditionally independent.

Takeaway

A graphical model turns **background knowledge into a factorisation**: missing edges are conditional independence assumptions, and each one removes parameters. The graph is a picture of how the joint splits — nothing more, and nothing less.